



Quantification of Nutrient Related Ecosystem Functions in *Zostera Marina* Meadows - A Review

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Abstract

Populations of seagrasses such as eelgrass (*Zostera marina*) have been declining worldwide within the last century. This loss has partly been attributed to mechanisms associated with increases in nutrient loading from anthropogenic sources. Furthermore, the systems are kept in a degraded state through feedback mechanisms caused by loss of the eelgrass habitat's nutrient-related ecosystem functions. In this article, we reviewed the available published data and methods used for the quantification of nutrient-related ecosystem functions. We focused on three pathways of nutrient retention/export: (1) Assimilation in the eelgrass biomass, (2) nutrient burial and sedimentary stocks, and (3) microbially driven nutrient cycling. We conceptualized the mechanisms driving these ecosystem functions and evaluated the current methods' applicability regarding the quantification of nutrient budgets. We found that studies often lack data from a comparable control site, and the ecosystem function could therefore not be derived. This was evident in studies of sediment stocks and microbial cycling, which often only included data from eelgrass. Standing biomass of eelgrass was commonly measured, but few considered the production and turnover of biomass as an annual cycle. Studies relating to microbial nitrogen cycling were the most abundant in the literature, but they often lacked simultaneous measurements of N-fixation and denitrification and thus did not allow for a net N₂-flux calculation. We concluded that there is need for studies that consider the underlying mechanisms driving the ecosystem functions across all topics, and that restoration sites are a great opportunity to investigate the development of the ecosystem functions that seagrass habitats facilitate.

Keywords *Zostera Marina* · Ecosystem Functions · Seagrass · Estuaries · Nutrient Cycling

Introduction

Seagrasses are the primary submerged rooted vegetation in many coastal ecosystems, with temperate seagrass communities consisting of a few or a single species, where the most common species in the northern temperate hemisphere is eelgrass, *Zostera marina* (Larkum et al., 2006). Large populations of eelgrass were lost throughout the Atlantic zone in the 1930s due to a wasting disease, from which much recovered during the following decades (Cotton, 1933; Giesen et al., 1990; Muehlstein, 1989). However, increasing population size and the industrialization of agricultural production in the latter half of the 20th century caused a rise in anthropogenic pressures on marine habitats and eutrophication. Consequently, the extent of eelgrass and other seagrasses was drastically reduced again as excessive nutrient loading from land shifted the ecosystems into an algae-dominated state (Burkholder et al., 1992; de Los Santos et al., 2019; Orth et al., 2006). This has left many waterbodies degraded and

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in poor ecological condition. Reductions in nutrient loading are the first necessary step towards ecosystem recovery. But, in cases where natural reestablishment of seagrass is absent post nutrient reductions, lack of seagrass coverage can be the bottleneck for obtaining better environmental conditions, as seagrasses perform a suite of essential biological functions, which facilitate the ecosystem recovery process (Carlos M. Duarte et al. 2013a, b; Lillebø et al. 2011; McGlathery et al. 2012). In this context, the ecosystem functions related to nutrient sequestration and reduced nutrient turnover are particularly important to consider. These functions are biomass-dependent, as they are controlled by the growth, productivity and recovery of the seagrasses (Flindt et al., 2024).

Today, it is recognized that in many systems the degradation is no longer directly reversible through reduction of the initial causing factor, e.g. excessive nutrient loading (Maxwell et al., 2017; Riemann et al., 2016; Unsworth et al., 2024). Studies point towards the retention of a degraded ecosystem state by ecological feedback mechanisms caused by a lack of ecosystem functions and resilience provided by essential habitats like eelgrass, such as nutrient buffering (Austin et al., 2017; Carstensen et al., 2012; Carlos M. Duarte et al. 2013a, b). When present, eelgrass affects most pathways of nutrient export/retention in coastal ecosystems and acts as a buffer to excess nutrients in the system (Cabaço et al., 2013; McGlathery et al., 2007). Without this buffer capacity, systems are more sensitive to nutrient-induced hysteresis, with positive feedback loops keeping them in a perpetually degraded state (Duarte et al., 2008), and reductions in external nutrient inputs will not necessarily bring waterbodies toward a greater ecological condition without considering the loss of essential habitats (Carstensen et al., 2012; Pedersen et al., 2013).

Examples show that despite major reductions (>50%) in N discharge to coastal waters, seagrass recovery can still be limited (Greening et al., 2011; Krause-Jensen et al., 2021), as several eutrophication-derived feedback loops maintain multiple stressors that decrease light availability for eelgrass and limit recolonization (Banke et al., 2025; Kenworthy et al., 2014; Krause-Jensen et al., 2005). First, sediment conditions with high organic content and no eelgrass cover maintain poor water clarity, as fine particles are easily resuspended and now contribute a significant amount to the turbidity of the waters (Austin et al., 2017; Marin-Diaz et al., 2019; Riemann et al., 2016). The fine particles also decrease the anchoring capacity for new eelgrass shoots, thus limiting natural recovery (Lillebø et al., 2011). Secondly, nutrients that would previously be bound in eelgrass are now available to phytoplankton and fast-growing macroalgae that bloom, collapse, and decay, resulting in rapid nutrient turnover and exacerbating hypoxic conditions (Conley et al., 2011; Flindt et al., 1999). These algae communities

also limit light availability for potential eelgrass recovery as phytoplankton increase light attenuation in the surface layer, opportunistic macroalgae cover the seafloor and epiphytes directly shade and smother eelgrass on its surface (Brodersen et al., 2015; Nielsen et al., 2002). Thirdly, a trophic cascade caused by bottom-up stimulation from eutrophication and top-down pressure from overfishing results in mesopredators thriving and feeding excessively on mesograzers that would otherwise assist in grazing away the algae and epiphytes, which shade and smother the eelgrass (Baden et al., 2012; Thormar et al., 2016).

A lack of natural eelgrass recovery, even after nutrient discharge reduction, has resulted in many new restoration initiatives during the past decades with variable outcomes (Flindt et al., 2024; van Katwijk et al., 2015). However, multiple projects have successfully restored eelgrass with some ecosystem functions returning almost immediately, while others depend on years of stable dense meadows (Lee & Park, 2008; Orth et al., 2020; Sievers et al., 2025). As the implementation of the new EU nature restoration legislation will encourage further restoration, and with eelgrass being a biological indicator of the ecological water quality in the European Water Framework Directive (EU, 2024; Krause-Jensen et al., 2005), it is essential to have a monitoring practice with quantifiable performance metrics, which measure ecosystem functions and evaluate success (Gann et al., 2019).

The published literature reveals that there is no clear consensus about the applied methods used for quantification of ecosystem functions related to nitrogen (N) and phosphorus (P) cycling in eelgrass meadows. Here, we aim to conceptualize these nutrient-related ecosystem functions in eelgrass, review the quantified and published results from the literature, present the variety of methods for quantification and discuss their applicability. We will use this to conceptualize the consequences of lost ecosystem functions in eutrophic systems, the lack of eelgrass coverage's effect on the recovery trajectory of degraded coastal habitats, and eelgrass' potential as a restoration measure. We focus on three major pathways of nutrient cycling:

- (1) Nutrient assimilation in the eelgrass biomass and the associated temporary immobilization of nutrients.
- (2) Nutrient burial and sedimentary stocks which may permanently immobilize nutrients within the system.
- (3) Microbially driven nutrient cycling, which imports and exports nutrients to and from the system.

The term immobilization in this review will be used as a blanket term for nutrients being biologically unavailable to other primary producers, whether temporarily or permanently.

Methods & Data Acquisition

Search Criteria, Grouping of Data

This review focuses on eelgrass (*Zostera marina*), as it is a foundation species in the coastal zone of the northern hemisphere (Larkum et al., 2006) and is the most studied seagrass species globally. We conducted a literature search using the Web of Science search engine for peer-reviewed literature, to query current published information on nutrient cycling in eelgrass meadows. The initial search was conducted in November 2024 and revisited in November 2025 to check for new additions to the literature pool, using the terms in Table 1. This generated 492 studies; of these, only 46 met the two criteria to be included in this study: (1) The study must describe specifics relating only to the *Z. marina* species

Table 1 List of search terms used in the Web of Science search engine, the number of hits provided and the subsequent studies that met the criteria of the search. Some studies showed up across multiple searches

Topic	Search terms	No. of studies	No. of studies that fulfill criteria
Nitrogen	<i>Zostera marina</i> ; nitrogen	10	40
	sequestering	66	
	<i>Zostera marina</i> ; nitrogen	21	
	accumulation	58	
	<i>Zostera marina</i> ; nitrogen	40	
	burial	46	
	<i>Zostera marina</i> ; nitrogen	122	
	sink	63	
	<i>Zostera marina</i> ; nitrogen		
	stock		
	<i>Zostera marina</i> ; nitrogen		
	storage		
	<i>Zostera marina</i> ; nitrogen		
Phosphorus	fixation		6
	<i>Zostera marina</i> ;		
	denitrification		
	<i>Zostera marina</i> ; phosphate	0	
	sequestering	7	
	<i>Zostera marina</i> ; phosphate	1	
	accumulation	6	
	<i>Zostera marina</i> ; phosphate	4	
	burial	15	
	<i>Zostera marina</i> ; phospho-	4	
	rus burial	17	
	<i>Zostera marina</i> ; phosphate	7	
	stock	5	
	<i>Zostera marina</i> ; phospho-		
	rus stock		
	<i>Zostera marina</i> ; phosphate		
	storage		
	<i>Zostera marina</i> ; phospho-		
	rus storage		
	<i>Zostera marina</i> ; phospho-		
	rus adsorption		
	<i>Zostera marina</i> ; phosphate		
	adsorption		

and (2) contain quantitative information on rates/stocks of either nitrogen or phosphorus related specifically to eelgrass (Table 1). Studies performed in both natural and restored habitats were included. The literature review was not limited to the results from the search strings. Other relevant studies that were cited in the literature from the search string were included to provide as comprehensible a dataset as possible.

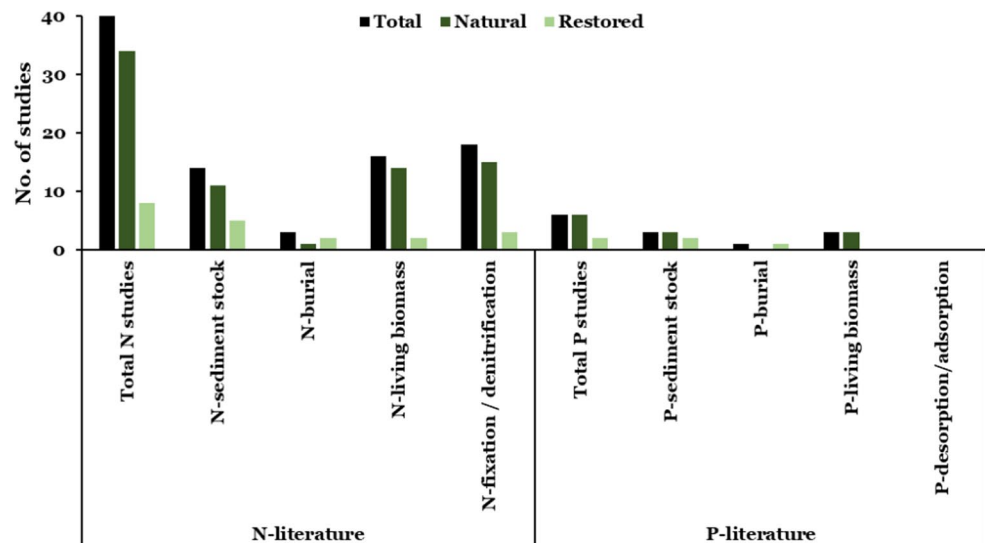
Data Availability and Quality

The studies included in this review originated from across the globe, spanning four continents: Europe, North America, Africa, and Asia, with the largest contribution from Europe and North America (Figure S1). A few studies provided data on restored eelgrass habitats and originated from only two countries: Denmark and the USA (Figure S1). The temporal spread of the studies was from 1973 to 2024. The number of studies per decade was only around 1–6 from the 1970's until the 2000's. However, this almost doubled in the 2010's and seems to double again based on the first half of the 2020's. The distribution of the results was skewed towards N-studies more so than P-studies with 40 and 6 studies, respectively (Fig. 1). Within the topical bounds of N, studies on the microbial processes of denitrification and N-fixation had the greatest representation followed by biomass studies, sediment stock and lastly burial (Fig. 1). Due to the scarcity of studies regarding P, the distribution was even across topics, except P de-/adsorption, where no studies were found (Fig. 1).

For calculations of the different ecosystem functions only a subset of the data extracted via the search was viable for each of the three ecosystem functions, and the values were not always reported directly in text and had to be extracted from figures. For nutrient assimilation in the biomass, only summer values were used to calculate the nutrient stocks in the standing biomass, as these were the most commonly available. For sedimentary nutrient stocks and microbial cycling, only sites with both eelgrass and unvegetated control samples were used to calculate the ecosystem function. The data used for calculations of N ecosystem functions is presented in tables S3.a (assimilation in the biomass), 2 (sediment stocks), and 3 (microbial cycling), and P ecosystem functions in table S2.b (sediment stocks). Here, it is also stated how the data was retrieved.

When calculating the different ecosystem functions, we needed to account for multiple values that collectively represent the ecosystem function. In the case of assimilation in the biomass, above- and belowground biomasses were combined, while ecosystem functions related to sediment stocks and microbial cycling were calculated as the difference between vegetated and unvegetated samples. In either case, we used the following equation to combine the standard errors:

Fig. 1 Summary of the number of studies. The studies are divided into different fractions or processes related to nutrient cycling. Black bars = Total number of studies, dark green bars = number of studies relating to natural eelgrass habitats, light green bars = number of studies relating to restored eelgrass habitats



$$SE_{total} = \sqrt{SE_a^2 + SE_b^2} \quad (1)$$

We then calculated the simple average of the means, i.e. the ecosystem function and used error propagation to calculate the errors:

$$SE_{propagated} = \frac{\sqrt{\sum SE_i^2}}{n^2} \quad (2)$$

Thus, all ecosystem functions are reported as the mean of means $\pm$ the propagated SE. In cases where standard deviation (SD) was reported instead of SE we calculated the SE from the number of replicates. In some cases, neither SD/SE was available in the reported data (Tables 2 and 3; Table S2.b; Table S3.b). In these instances, we used the median SE from the remaining data to calculate the propagated.

Statistical analysis using a one-tailed and paired t-test in Microsoft Excel was performed to test if sediment N stocks were greater in eelgrass sediments compared to unvegetated controls (as well as the net N_2 -flux from microbial cycling in eelgrass sediments being greater compared to unvegetated controls. This analysis was only performed on the data presented in Tables 2 and 3.

Results

Nutrient Assimilation in the Biomass

It is well documented that eelgrass assimilates dissolved inorganic N and P through both its roots and leaves, affecting nutrient concentrations in both the sediment and the water column (Pedersen & Borum, 1992; Short & McRoy, 1984). Thus, the standing biomass of eelgrass constitutes a

constant stock of immobilized nutrients in the system, and any increase in biomass through areal expansion constitutes additional immobilization. Furthermore, seasonal variation in eelgrass biomass, both above- and belowground, causes a temporary nutrient immobilization during the growth season when production peaks. Then, as the growth season ends, a significant amount of biomass is lost deciduously, some of which is degraded within the system, thereby returning the nutrients (Fig. 2) (Moore, 2004; Olesen & Sand-Jensen, 1994; Risgaard-Petersen & Ottosen, 2000). The biomass which is lost during and after the growth season from sloughing or senescence creates a detrital stock of nutrients which has a slow turnover rate (100–700 d) due to the presence of lignin and other recalcitrant molecules which slow degradation, and most nutrients will not be released within the same growth season (Banta et al., 2004; Trevathan-Tackett et al., 2020). Before sloughing its leaves, eelgrass reabsorbs some of the nutrients from the leaves (Hemminga et al., 1999). This is contrary to the fast turnover of competing phytoplankton and opportunistic macroalgae, with a boom-and-bust life strategy and easily decomposed biomass that allows for multiple pulses of nutrients to reenter the system within the same growth season, an effect which is amplified in eutrophic systems (Banta et al., 2004; McGlathery et al., 2007). In addition, eelgrass leaves tend to float due to the aerenchyma in the tissue, which leads to some export from the system onto land or out to deeper waters (Duarte & Krause-Jensen, 2017; Flindt et al., 2004). As a result, nutrients are circulated fewer times in eelgrass-dominated systems, hereby limiting available nutrients that would otherwise support the growth of phytoplankton and macroalgae and buffering the nutrient-dependent hysteresis of the system (Flindt et al., 1999; McGlathery et al., 2007).

There is a distinct seasonality to the immobilization of nutrients in eelgrass biomass throughout an annual cycle,

Table 2 Literature values for total-N-stock in eelgrass meadow sediments and a comparable unvegetated. Literature values without controls are not shown. The living biomass of eelgrass has been removed from the sediment before analysis unless otherwise stated. Only values measured in both natural and restored meadows along with a unvegetated control are included. In measurements regarding restored meadows the age of the restoration is stated. Other information stated are the location name, the depths of the cores (cm) and replicate count (n) in cases with different number of replicates for each treatment eelgrass is stated first, then unvegetated, and source of information. Dashed lines indicate separation between studies

Natural eelgrass					
Country/Area	N-stock (g m ⁻²)		Core depth and count	Source	
	Eelgrass	Unvegetated (n)			
Denmark					
Horsens Fjord	55.9±4.3 SE	34.6±2.2 SE	15 cm, 6 n	Lange et al., 2022	
Horsens Fjord	45.6±4.07 SE	51.9±8.14 SE	15 cm, 3 n	Steinfurth et al., 2022	
Horsens (Protected)	171±40 SD ^U	119±10 SD ^U	50 cm, 3 n	Leiva-Duenas et al. 2023	
Horsens (Exposed)	152±52 SD ^U	287±29 SD ^U			
Nibe (Protected)	169±20 SD ^U	191±7 SD ^U			
Nibe (Exposed)	348±25 SD ^U	214±8 SD ^U			
Syd Fyn (Protected)	207±13 SD ^U	370±95 SD ^U			
Syd Fyn (Exposed)	174±52 SD ^U	131±11 SD ^U			
Roskilde (Protected)	225±35 SD ^U	170±13 SD ^U			
Roskilde (Exposed)	357±63 SD ^U	372±17 SD ^U			
Sweden					
Gullmarsfjord (Protected)	2011.4 ^Y	1714.1 ^Y	120 cm, 3 n	Moksnes et al. 2021a, b	
USA					
North Harbour	93±10 SD ^U	30±6 SD ^U	5 cm, 6 n	Prystay et al., 2023	
Swift Current	85±6 SD ^U	69±10 SD ^U	5 cm, 6 n		
Baie de l'Eau	95±20 SD ^U	120±30 SD ^U	5 cm, 6 n		
Restored eelgrass					
Country/Area	Age (year)	N-stock (g m ⁻²)		Core depth and count	Source
		Eelgrass	Unvegetated		
Denmark					
Horsens Fjord	2	49.4±2.0 SE	34.6±2.2 SE	15 cm, 6 n	Lange et al., 2022
Horsens Fjord	2	57.3±3.7 SE	51.9±8.1 SE	15 cm, 3 n	Steinfurth et al., 2022
USA					
Hog Island Bay	4	28.8 ^a	30.6 ^a	10 cm, 24 and 23 n	Greiner et al., 2013
South Bay	10	65 ^a	32.2 ^a	10 cm, 24 and 8 n	
South Bay	9	16.5±1 SE*	5±0 SE*	5 cm, 5 n	McGlathery et al., 2012
Hog Island Bay, South Bay, Cobb Bay, Spider Crab Bay	1–12	19.4 ^{a, b, U, Y}		5 cm, 4 n	Orth et al., 2020

* Value read of graph in the original source.

^a Value calculated based on sediment characteristics provided in original source. SD or SE not possible to present due to lack of raw data.

^b Value represents the difference between sediment N-stock in eelgrass and unvegetated.

^U Unknown if living root biomass is included in the stock

^I Value based on a single sediment core, no replicates.

^Y SE or SD not available

* Value read of graph in the original source.

^a Value calculated based on sediment characteristics provided in original source. SD or SE not possible to present due to lack of raw data.

^b Value represents the difference between sediment N-stock in eelgrass and unvegetated.

^U Unknown if living root biomass is included in the stock

^I Value based on a single sediment core, no replicates.

^Y SE or SD not available

whether it is in a temperate climate (Risgaard-Petersen & Ottosen, 2000; Sand-Jensen, 1975), or towards the southern limit of the distribution, where the peak biomass is earlier in spring (Moore, 2004) (Fig. 2). Eelgrass populations closer to the equator will of course experience warmer temperatures and more hours of substantial sunlight penetration due to more hours with higher solar angles ($>10^\circ$) (Weinberg, 1976), and where the northern populations lose biomass after the growth season due to sunlight limitation, the southern populations may lose it due to build up of detritus and

macroalgae and increases in water turbidity instead (Moore, 2004). However, starting from the minima of biomass, both growth cycles are similar. Here, we use the northern distribution as an example due to the greater expression of seasonality: In spring, biomass production starts to increase as the standing stock grows with leaves getting longer and thicker, the number of leaves per shoot increases and the density of shoots increases (Fig. 2). In the next season of summer, both productivity and standing biomass are at their maximum, and the rapid turnover of leaves causes sloughing

Table 3 Literature values for the microbially driven processes N-fixation (N-fix) and denitrification (Denit). Values are only presented from studies that include both N-fixation and denitrification or directly measure Net N₂-flux. All values are measured at ambient N-levels. Values for N-fixation are negative, indicating an input of N to the system, while denitrification exports N from the system. Only values measured in natural and restored meadows along with a unvegetated control, are included. When studies measure in restored meadows, the age of the restoration is stated. Other information stated is the location name, measured medium, method used, and source of information. Dashed lines indicate separation between studies

Natural eelgrass								
Country/Area		N ₂ -flux (mg N m ⁻² d ⁻¹)		Measured medium	Method	Source		
		Eelgrass	Unvegetated					
Denmark								
Roskilde Fjord		21 ^{h, y}	12.32 ^{h, y}	Rhizosphere + Phyllosphere	Mass- balance study	Flindt, 1994		
USA								
Cormorant Point		-2.69±2.69 SE*, a, h	21.50±10.75 SE*, a, h	Rhizosphere + Phyllosphere	N ₂ -flux, <i>ex situ</i>	Zarnoch et al., 2017		
Tiana Bay		8.4±5.71 SE*, a, h	-3.36±10.08 SE*, a, h					
Tiana Beach		9.41±10.08 SE*, a, h	-9.07±1.34 SE*, a, h					
Chesapeake Bay (Spring)		23.3±5.83 SE*, a, h	-17.5±8.75 SE*, a, h	Rhizosphere + Phyllosphere	N ₂ -flux, <i>ex situ</i>	French et al., 2024		
Chesapeake Bay (Summer)		35±5.83 SE*, a, h	0.58±10.50 SE*, a, h					
Tiana Beach (Ambient pH and N)		-58.33*, a, h, m, y	-87.5*, a, h, m, y	Rhizosphere + Phyllosphere	N ₂ -flux, <i>ex situ</i>	Kahn et al., 2023		
Restored eelgrass								
Country/Area	Age (year)	N ₂ -flux (mg N m ⁻² d ⁻¹)		Measured medium		Method	Source	
		Eelgrass		Unvegetated				
		N-fix	Denit	N-fix	Denit			
USA								
South Bay	13–16	-2.55±0.17 95%CL ^a	1.70±0.21 95%CL ^a	-2.03±0.44 95%CL ^a	0.44±0.14 95%CL ^a	Rhizosphere + Phyllosphere	Acetylene reduc- tion+ ¹⁵ NO ₃ Push-pull in situ	Aoki et al., 2020
Hog Island Bay	3	10.65±2.55SE ^{a, h}		3.23±4.91SE ^{a, h}		Rhizosphere	N ₂ -flux, <i>ex</i>	Cole,
South Bay	8	15.89±4.27SE ^{a, h}		3.23±4.91SE ^{a, h}			<i>situ</i>	2011
* Value read of graph in original source								
^a Value calculated based on daily- (mg d ⁻¹) or yearly rates (mg yr ⁻¹) provided in the original source.								
^h Net N ₂ -flux. Net rate of both N-fixation and denitrification, which when positive is a net removal of N from the system, and when negative is a net introduction to the system.								
^m Value is median read of graph.								
^y SE or SD not available								

* Value read of graph in original source

^a Value calculated based on daily- (mg d⁻¹) or yearly rates (mg yr⁻¹) provided in the original source.

^h Net N₂-flux. Net rate of both N-fixation and denitrification, which when positive is a net removal of N from the system, and when negative is a net introduction to the system.

^m Value is median read of graph.

^y SE or SD not available

(Fig. 2). After this peak season, productivity decreases in autumn due to fading light conditions, and standing biomass decreases as sloughing is high (Fig. 2). Finally, in winter, productivity and sloughing are minimized, and the standing biomass is at its lowest, which represents the constant stock of immobilized nutrients (Fig. 2). Thus, every year a large amount of eelgrass biomass is produced and subsequently sloughed, which assimilates and temporarily immobilizes nutrients (100–700 d) because of the slow turnover rate of eelgrass tissues (Banta et al., 2004). As a result, the sum of this production is greater than simply the standing stock at a given time and is an ecosystem function that is often overlooked. This temporary immobilization can be conceptualized similarly to that of a lung withdrawing nutrients at the most ecologically crucial time and releasing them in a

period of lesser impact (Fig. 2). This is often missed in biomass studies as the focus remains on the standing stock of biomass, whereas net primary production is a representation of long-term burial in the system (Kristensen et al., 2025). On the contrary, gross annual biomass production may be a good representation of the temporary nutrient immobilization (Fig. 2). Some of these nutrients will be mineralized back into the system during the growth season. But most will be mineralized following the end of the growth season, due to the slow degradation rates (Banta et al., 2004).

Multiple studies have investigated the productivity of eelgrass, most commonly using a variation of the leaf-marking method (Ibarra-Obando & Bourdoursque, 1994; Kentula & McIntire, 1986; Short & Duarte, 2001). But only a few studies manage to monitor the annual production cycle

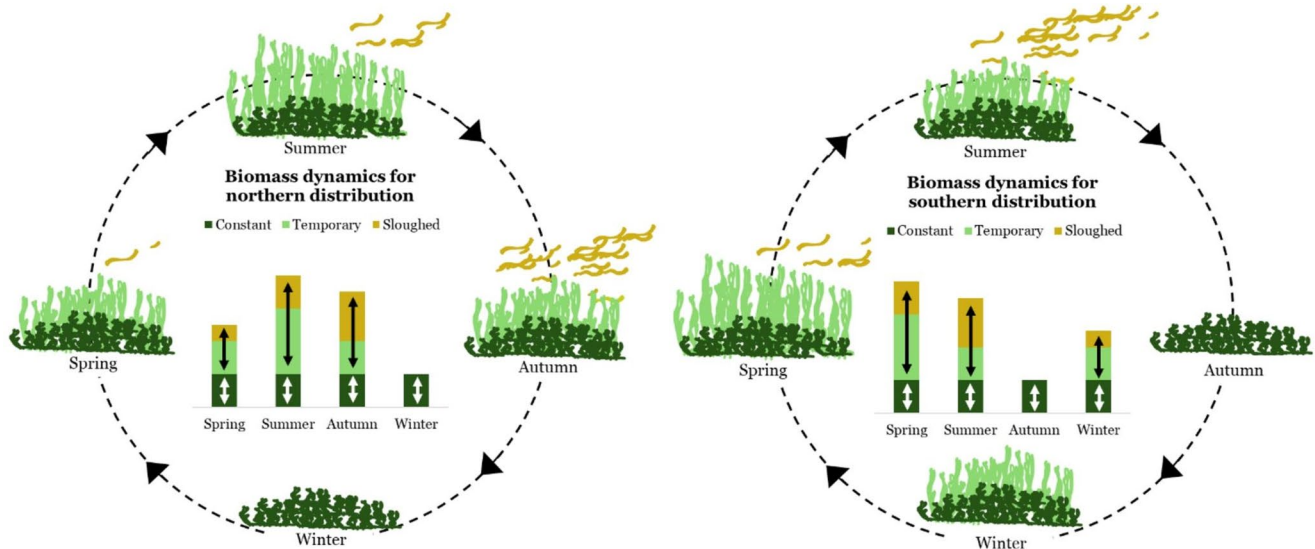


Fig. 2 Conceptual diagram showing an annual cycle of constant (dark green), temporary (light green), and sloughed (orange) eelgrass biomass. As well as the ecosystem function of temporary (black double headed arrows) and constant (white double headed arrows) immo-

bilization of nutrients in the biomass. Left panel: Seasonal biomass dynamics for the northern distribution of *Z. marina*. Right panel: Seasonal biomass dynamics for the southern distribution of *Z. marina*

of eelgrass. An American study quantified the annual assimilation of N in eelgrass biomass based on bimonthly observations of biomass production during the growth season (Apr. – Oct.), which amounted to $2.62 \text{ g N m}^{-2} \text{ yr}^{-1}$ (Aoki et al., 2020). For comparison, a Danish study also quantified eelgrass production during the growth season (Apr. 9 – Oct. 16) and found a total gross production of 1097 g dw m^{-2} , corresponding to 2.5 times the maximum total standing biomass in peak summer (Sand-Jensen, 1975). Based on the means of N content in the leaves, rhizomes and roots across the literature (Table S4), this results in a total uptake during the growth season of 15.0 g N m^{-2} , indicating both the magnitude and large variability of this ecosystem function.

This variability is also evident in the literature on standing biomass, as N-stocks in the total living biomass range from 1.5 to 9.8 g N m^{-2} with a mean of $5.2 \pm 0.2 \text{ g N m}^{-2}$ (mean $\pm$ propagated SE) (Fig. 3). Measurements of P were performed in only 3 studies and ranged from 0.3 to 0.8 g P m^{-2} in the total biomass (Table S3.b). Monthly variation of the N-stocks in standing eelgrass biomass throughout the seasons was investigated in only two studies, one in the USA and one in Denmark (Moore, 2004; Risgaard-Petersen & Ottosen, 2000). In the temperate climate of Denmark, the leaf biomass reaches its maximum during summer and minimum during winter, with N-stocks ranging from 1.1 to 11.2 g N m^{-2} (Risgaard-Petersen & Ottosen, 2000). In the warmer climate of the Chesapeake Bay in the USA, the biomass reaches a maximum during early spring, then dissipates during the summer, with N-stocks in the leaves ranging from 0.5 to 5.9 g N m^{-2} (Moore, 2004). C and N contents in

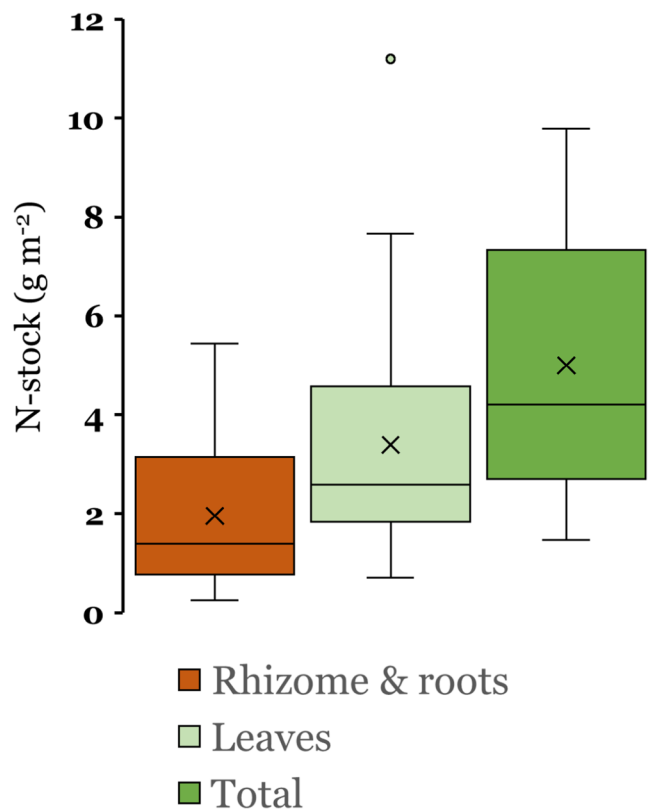


Fig. 3 Literature values for N-stocks in the living biomass of eelgrass. Values are presented for roots and rhizomes ($n=26$), leaves ($n=32$), and total biomass ($n=27$). The cross (X) is the mean of the data, the center horizontal line is the median, the box represents the interquartile range between the 1st and 3rd quartile of data, and error bars are the minimum and maximum values of the data, excluding outliers. Sources and data can be found in Table S3

the living biomass and their ratio (C: N) also change seasonally, as well as in response to nutrient availability (Guidetti et al., 2001; Moore, 2004; Schmidt et al., 2012). C and N content in the biomass are, on average, 32.4% C and 1.5% N of DW in aboveground biomass and 31.0% C and 0.9% N of DW in the belowground biomass (Table S4). Eutrophication can lead to increases in N content of around 0.5% of DW, but this is accompanied by lower shoot densities and area-specific biomass. Consequently, the area-specific N-stocks do not increase (Schmidt et al., 2012). Seasonality is more important as N contents may vary by > 1% of DW between summer and winter values (Guidetti et al., 2001). As the growth season initiates, the plants utilize their stored N content and the C: N increases and peaks during maximum standing biomass. However, once light is limited and growth minimizes, N again becomes available as supply from degradation and external loading exceeds growth related assimilation rates, and the N stores are replenished (Guidetti et al., 2001; Moore, 2004). The N content of the sloughed leaves is rarely investigated but has been shown to range between 1 and 2% N of DW throughout the degradation process, and vary depending on temperature and oxygen conditions (Godshalk & Wetzel, 1978). There is little available data on P contents in the biomass, but in the few cases where it is measured, the variance is high (Moore, 2004). It is unclear whether nutrient reabsorption also varies seasonally. Still, this is important knowledge to gain, as this will affect the amount of nutrients reentering the system.

Nutrient Burial and Sedimentary Stocks

The presence of eelgrass meadows can increase nutrient stocks in most sediments through several mechanisms. Firstly, the direct deposition of the eelgrass biomass itself as roots, rhizomes, and some leaf fragments leads to accretion of organic matter and associated nutrient content (Schmidt et al., 2011). Secondly, eelgrass alters the relationship between erosion and sedimentation. The stabilizing effects of the rhizosphere drastically reduce erosion of the sediments (Carlos M. Duarte et al. 2013a, b; Marin-Diaz et al. 2019), and the leaf canopy causes wave attenuation and reduction of currents, thereby increasing particle settling as resuspension decreases, resulting in an initial net increase in sedimentation (Hansen & Reidenbach, 2012; Infantes et al., 2022; Luhar et al., 2017). Studies on restored habitats show that accumulation rates are positively correlated with the age of the restored meadows, which is credited to the canopy structure becoming denser (Aoki et al., 2020). However, some literature assumes a continuous accumulation of nutrients in eelgrass meadows; this is especially evident in the growing number of studies focusing on blue-carbon effects (Greiner et al., 2013; Krause-Jensen et al., 2022). Other studies have

challenged this assumption, highlighting that the accretion of bulk sediment, and, in turn, the accumulation of nutrients, in eelgrass most likely only occurs until a new steady-state has been reached in relation to the surroundings (Kristensen et al., 2025). As eelgrass is introduced and the canopy becomes denser, sediment accumulation and the concentrations of N and P in the sediment increase to finite levels that determine the upper limit for the potential stock of N and P in the sediment. Then, over time, the concentration of nutrients in the sediment is balanced with mineralization and denitrification rates, which will increase as more labile organic compounds become available. Likewise, bulk sediment accretion is balanced by compaction of the sediment as well as erosion and resuspension, which will increase as bulk density and water depth decrease. When balanced, this results in N and P stocks in eelgrass sediments being greater compared to unvegetated sites (Fig. 4).

Since sediment stocks depend on canopy density, it is reasonable to assume that sediment accretion rates will exhibit a seasonal component. In a temperate climate, sediment accretion should be high in the summer, when the weather is calm, and shoot density is high. On the contrary, during winter, sediment accretion should be low or even negative due to rougher weather and a sparse canopy. In areas overall prone to net sedimentation due to local hydrology, such as depressions or embayments, continuous sediment accumulation is more likely to occur. In such areas, where eelgrass might reside, net sediment accumulation is likely in both eelgrass meadows and unvegetated areas. Thus, this highlights the importance of unvegetated controls to derive the ecosystem function of the eelgrass as this otherwise could be greatly overestimated. As an example, at a 9-year-old American eelgrass restoration site, they measured a sediment accretion of $\sim 0.6 \text{ cm yr}^{-1}$ where the unvegetated control (before restoration) had a sediment accretion of $\sim 0.1 \text{ cm yr}^{-1}$ (Greiner et al., 2013), in 100 year this would amount to a difference in sediment height of 50 cm between the unvegetated and eelgrass sediments. But at some point, eroding forces will likely increase due to the difference in sediment height to such an extent that accretion within the meadow will stop, and a balance will be achieved. The time for this new balance/equilibrium to be reached and for the ecosystem function to be fully developed remains undescribed in the literature and likely varies drastically with local environmental conditions. Likewise, the sediment depth to which the nutrient stocks are enhanced will vary from site to site.

The dynamic nature of coastal environments makes measuring net sedimentation challenging (Gallop et al., 2015; Kristensen et al., 2025). Direct measurements of N-accumulation rates in eelgrass meadows are only present in four studies: two performed in natural eelgrass meadows (Dahl et al., 2024; Leiva-Dueñas et al., 2023), and two in restored

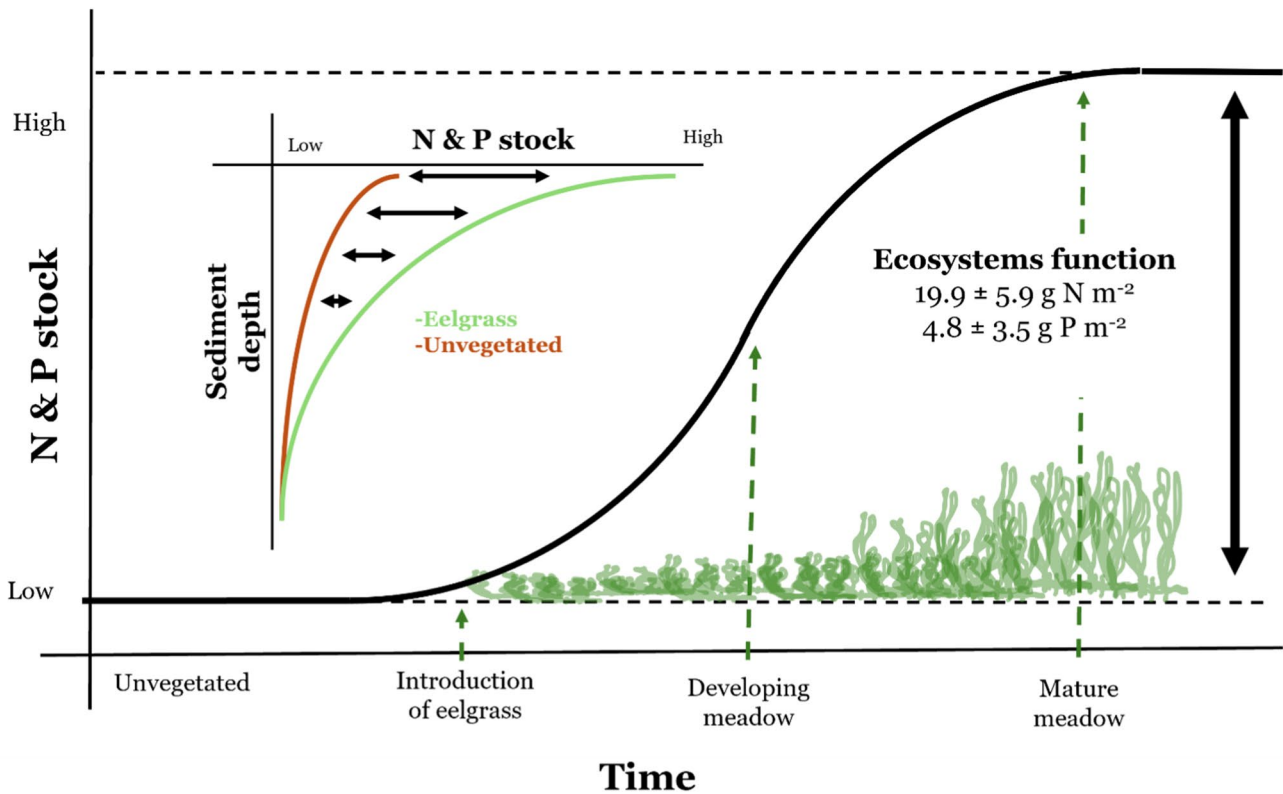


Fig. 4 Main plot: Conceptual diagram of the development of sediment stocks of N and P stocks (black line) going from unvegetated to a mature eelgrass meadow. Sub plot: N and P stocks and sediment depth for eelgrass (light green line) and unvegetated controls (red line). The

ecosystem function (solid black double headed arrows) is the mean of means ± propagated SE of the difference between eelgrass and unvegetated controls (N - $n=20$, P $n=4$)

meadows (Aoki et al., 2020; Lange et al., 2022). In three of the studies, they calculated the N-accumulation rate using the ^{210}Pb method (Aoki et al., 2020; Dahl et al., 2024; Leiva-Dueñas et al., 2023), a method commonly used for measuring sedimentation of fine particles (Barsanti et al., 2020). However, the method is severely vulnerable to sediment transport, bioturbation, and episodic erosion and vertical mixing due to storms (Barsanti et al., 2020; Gallop et al., 2015; Kristensen et al., 2025), which are all common in coastal sediments, and for one of the studies, failed to provide data for 5 out of 8 stations (Leiva-Dueñas et al., 2023). The N-accumulation rates at the remaining 3 stations ranged from 0.5 to 14 g N m⁻² yr⁻¹. However, no unvegetated control was provided, so the effect of eelgrass cannot be derived from the results (Leiva-Dueñas et al., 2023). For the Swedish study, two 2 m sediment cores were sampled within eelgrass meadows that had existed for >1000 year, using not only ^{210}Pb but also ^{14}C -dating (Dahl et al., 2024). Here, the rates within the old and sheltered meadows were 0.50–0.75 g N m⁻² yr⁻¹ (Dahl et al., 2024), but again, no unvegetated control was sampled, so the effect of eelgrass cannot be derived. N-accumulation was also measured in restored eelgrass meadows in Denmark and the USA (Aoki et al., 2020; Lange et al., 2022). In the American study, the ^{210}Pb methodology was also used (Aoki

et al., 2020). In contrast, the Danish study calculated the accumulation rates using the difference in N-stock between an unvegetated control and the restored meadow, as well as the restored meadow's age (Lange et al., 2022). As the only study in this review, the Danish study derived both N- and P-accumulation rates, with 6.6 g N m⁻² yr⁻¹ and 3 g P m⁻² yr⁻¹, respectively (Lange et al., 2022). Even though this was done in a 2-yr old bed, it is nearly twice the rate found in South Bay in the USA after a decade, 3.35 g N m⁻² yr⁻¹. This is likely due to increased eutrophication-related pelagic production in the eutrophic Danish systems and the growth of epiphytic algae on eelgrass leaves (Conley et al., 2000), thereby increasing the burial of organic particles (Aoki et al., 2020; Lange et al., 2022). However, in both cases, accumulation rates are high because the meadows are still maturing, which is evident from the American study, where sediment accretion continues to increase in the most recent measurements (Aoki et al., 2020), if measured over a longer time scale we expect the accumulation rate to be denominationally diluted by time.

Whereas measurements of N-accumulation rates are sparse in the literature, sediment stocks of N are quantified far more frequently, often as a byproduct of blue carbon studies (Dahl et al., 2016; de la Cerda-Marín et al.,

2023; Duarte et al., 2010). The data on sedimentary N stocks range from 16.5 to 2011.4 g N m⁻² in eelgrass and 5–1714.1 g N m⁻² on unvegetated sediments (Table S2.a). The immense variation in the data is partially caused by a lack of consensus on a standardized sediment core depth, demonstrated by the range of core depths, 5–120 cm, from studies included in this review (Table S2.a). Furthermore, this variation is enhanced by the natural heterogeneity of coastal sediments. Areas protected from hydrodynamic pressure often accumulate more organic matter and therefore, provide much greater values for nutrients as well as carbon stocks, both in and outside eelgrass meadows (Kindeberg et al., 2018; Leiva-Dueñas et al., 2023; Per-Olav Moksnes et al. 2021a, b).

For eelgrass meadows to provide a nutrient-burial ecosystem function, the sediment nutrient stocks of eelgrass meadows need to be greater than those of a comparable control site (i.e. unvegetated sediment). However, only 20/104 sites where nutrient stocks are measured in eelgrass included a comparable unvegetated control site (Fig. 3; Table S2.a), indicating a lag in studies suitable for assessing the nutrient-burial ecosystem function of eelgrass. Additionally, studies were often focused on carbon and included nutrient data only as a supplement. There was no statistically significant increase in sediment N stocks when comparing eelgrass to unvegetated controls across the 20 sites ($p=0.18$, $n=19$) (Table 2). However, on average, eelgrass meadows increased sediment stocks by 19.9 ± 5.9 g N m⁻² (mean $\pm$ propagated SE), with eelgrass sites stocks of 231.9 ± 3.8 g N m⁻² and unvegetated sites stocks of 212.0 ± 3.3 g N m⁻² (mean $\pm$ propagated SE) (Fig. 4). This difference is not a strongly expressed trend, as 7/20 sites show that nutrient stocks are greater in unvegetated sediment than in eelgrass meadows (Table 2). This emphasizes the importance and impact of control site selection.

Choosing an appropriate control site is inherently difficult in coastal environments due to the heterogeneity and mobility of sediments. In Denmark, two separate studies have been performed on the same restored eelgrass meadow (Lange et al., 2022; Steinfurth et al., 2022). In the first, the control was placed within a few meters of the restored site, and a significant increase in N-stock was found 2 years after the site was restored, with 34.6 ± 2.2 g N m⁻² (mean $\pm$ SE) and 49.4 ± 2.0 g N m⁻² (mean $\pm$ SE) for the control and restored site respectively (Lange et al., 2022) (Table 2). The second study had the control site placed > 100 m east of the restoration and was unable to show a significant difference due to variation in the control samples (Steinfurth et al., 2022) (Table 2). There is a natural heterogeneity in the sediments caused by random layers of peat, shells, and clay, as well as variable

grain-size composition, which can drastically alter results. Therefore, there should be a greater focus on increasing the number of replicates to properly cover the variation. Other studies also find greater N-stock at the unvegetated control site, contrary to expectations. This was often found in cases with high organic content due to sites being sheltered (Table 2) (Leiva-Dueñas et al., 2023; Prystay et al., 2023). However, since the mechanisms driving the ecosystem function are primarily physical, sheltered sites should be expected to produce less of a signal, as hydrodynamics are already at a minimum, and further wave and current attenuation provided by eelgrass canopy is unlikely to increase sediment stocks. Instead, the signal from the autochthonous contribution of deceased eelgrass biomass should still exist, which emphasizes the need for more eelgrass studies which include unvegetated controls and investigations into selecting an appropriate control. In this regard, restoration sites can provide a unique opportunity to perform Before-After-Control-Impact studies (BACI), as samples can be collected before restoration is initiated, during development, and after the restoration is fully developed, thus providing the greatest accuracy for the actual ecosystem functions and the time needed for the ecosystem functions to develop (Christie et al., 2019).

Only two studies included P-stock data with a comparable unvegetated control. However, both studies originated from the same restoration site in Denmark with an ecosystem function in the range 2.1–7.4 g P m⁻² (Table S2.b) (Lange et al., 2022; Steinfurth et al., 2022). Lack of research on the subject of P has been a commonality for over 30 years (Hemminga et al., 1991). We found no studies on the effect of eelgrass on P adsorption in coastal sediments. This is in spite of the fact that, in the early spring, P has been demonstrated to be the limiting nutrient for phytoplankton blooms in some temperate systems (Conley et al., 2000; Nielsen et al., 2002). Therefore, the P adsorption capacity in the oxygenated rhizospheres of eelgrass is important to quantify along with the sedimentary stocks. Though there is no data on eelgrass, sediments oxidized by dwarf eelgrass (*Zostera noltii*) have been found to increase the capacity for P-adsorption as oxidized iron (III)-compounds bind dissolved inorganic P (Deborde et al., 2008). As O₂-leakage is also well documented from eelgrass roots, it can be inferred that a similar increase in absorption capacity is likely to be present (Brodersen et al., 2015; Frederiksen & Glud, 2006). Similar to sediment accretion, the rate of P adsorption is likely not continuous and reaches a steady-state depending on meadow density. In a meadow at balance, the eelgrass is neither source nor sink. However, studies researching the increased internal loading of P in coastal ecosystems as a consequence of lost eelgrass coverage are relevant to understanding the recovery trajectories (Duarte et al., 2008).

Microbially Driven Nutrient Cycling

Several microbial processes contribute to the cycling of bioavailable N in estuaries and coastal waters, of which the largest sources and sinks are N-fixation and denitrification (McGlathery et al., 2007). By contrast, anammox (anaerobic ammonium oxidation) is generally not an important N sink in estuaries compared to denitrification (Damashek & Francis, 2017; McGlathery et al., 2007). Eelgrass can affect denitrification in several ways; in an oligotrophic system where N is scarce, denitrification may be inhibited through competition for NO_3^- as eelgrass roots can assimilate the available nutrients in the sediment (McGlathery et al., 2007; Risgaard-Petersen & Ottosen, 2000). In a eutrophic system, eelgrass can stimulate nitrification by introducing NH_4^+ and O_2 and denitrification by introducing dissolved organic carbon (French et al., 2024; McGlathery et al., 2007; Zarnoch et al., 2017). It is speculated that the presence of eelgrass also increases the coupled nitrification-denitrification, due to the O_2 leakage from the roots in the rhizosphere increasing the volume of sediment where coupled nitrification-denitrification can proceed (Itzumi et al., 1980). However, even though there is evidence that eelgrass stimulates both processes separately, there are no direct measurements of the coupled process being stimulated. It would, however, be further assisted by a richer burrowing infauna relative to unvegetated sites, where processes are otherwise restricted to the upper few millimeters of the sediment near the oxic-anoxic interface (Brodersen et al., 2024; French et al., 2024; Steinfurth et al., 2022).

Estuarine sediments harbor relatively low N-fixation rates relative to other benthic coastal ecosystems (Fulweiler et al., 2025), and the loss of N due to denitrification typically exceeds the input of N-fixation in estuaries (Fulweiler & Heiss, 2014; Seitzinger, 2003). The presence of eelgrass can stimulate N-fixation in both the phyllo- and rhizosphere (Brodersen et al., 2024; Marino et al., 2023), but N-fixation is energetically costly, and the enzyme responsible for the process, nitrogenase, has been found to be inhibited by the presence of dissolved inorganic N (Andersen et al., 2014). Therefore, it is only expected to dominate in environments where N is scarce, as is the case in *Posidonia oceanica* meadows in the Mediterranean (Agawin et al., 2017). As a result, the net effect of the opposing denitrification and N-fixation (the net N_2 -flux) is considered positively correlated to eutrophication (Fulweiler & Heiss, 2014), and the presence of eelgrass can further increase this efflux of N_2 in a eutrophic nutrient regime by stimulating denitrification (Fig. 6) (Zarnoch et al., 2017). But in an oligotrophic nutrient regime, the presence of eelgrass can cause a greater net influx of N_2 into the system, by hosting N-fixating bacteria (Fig. 6) (Agawin et al., 2017).

Denitrification and N-fixation are frequently investigated in the literature. Still, direct comparisons of the combined ecosystem function across studies are difficult due to variations in methodology (Table 3) (Newell et al., 2016). Furthermore, few studies include both opposing processes as well as unvegetated control sites, drastically limiting the ability to quantify the ecosystem function (Fig. 3). Historically, N-fixation has been investigated via acetylene reduction (Fulweiler et al., 2025; Herbert, 1999; Welsh, 2002). The enzyme nitrogenase reduces acetylene to ethylene, which is then measured as a proxy for N-fixation. Additionally, most studies focus only on the processes within the rhizosphere (Table S5), and while denitrification is limited to the rhizosphere, N-fixation has been demonstrated to occur in both the rhizosphere and phyllosphere (Marino et al., 2023; Welsh, 2002). The contribution of N-fixation by epiphytes in the phyllosphere is often overlooked in the literature, but can have a significant effect (Marino et al., 2023), though excessive epiphyte biomass will negatively affect the eelgrass blade itself. Denitrification has also been measured using acetylene, as it inhibits the last step from N_2O to N_2 in denitrification, and the accumulation of N_2O over time thus provides a rate of denitrification. This method was replaced in the 1990's by the isotope pairing technique (Nielsen, 1992). Alternatively, net fluxes of N_2 , typically determined by ratioing to fluxes of inert argon (N_2/Ar), provide the net outcome of the two opposing processes (Welsh, 2002; Zarnoch et al., 2017). The problem with these *ex situ* methods is that they only provide momentary measurements and suffer from the considerable change in porewater chemistry caused by sampling. During sampling, surface sediment may be pushed down and compacted, affecting oxygen conditions and eelgrass roots and rhizomes are severed, causing leakage of NO_x and dissolved organic carbon, and will affect rates of N-fixation and denitrification (Newell et al., 2016). For N-fixation, a new *in situ* method using $^{30}\text{N}_2$, called push-pull, has been developed and measures N-fixation that is an order of magnitude greater than in previous studies, primarily attributed to measurements being *in situ* (Aoki & McGlathery, 2019). Nevertheless, all methods are resource- and time-consuming, and fluxes are measured across a span of a few hours (2–6 h), which is inadequate to precisely estimate a mass balance budget. Generally, applying rate-based measurements to a highly complex rhizosphere is problematic when estimating the integrated effects of the ecosystem function. Instead, direct measurements of changes in system N-stocks can provide more precise estimates of mass-balance effects and act as a proxy for net N_2 -flux (Flindt, 1994).

In total, there were only six studies across ten sites from which the net denitrifying ecosystem function of eelgrass could be derived, four measured in natural meadows and

two in restored meadows (Table 3). The magnitude of microbial processes displays a high level of variation across studies (Table 3). The causes of the variation are hard to explain as there are multiple driving factors e.g. temperature, hydrodynamics, organic content of the sediment, and oxygen. The highest rates of net N_2 -efflux were measured in eelgrass meadows in Roskilde Fjord and Chesapeake Bay (Near the York River outlet), which are both considered affected by eutrophication. Here, net N_2 -efflux in eelgrass meadows exported 8.68 and $40.8 \text{ mg N m}^{-2} \text{ d}^{-1}$ more than the unvegetated control site, respectively (Flindt, 1994; French et al., 2024) (Table 3). By contrast, two studies performed on restored eelgrass meadows in the oligotrophic South Bay (Aoki et al., 2020; Cole, 2011). The 2011 study found a net N_2 -efflux in both the eelgrass and unvegetated sites with a difference of 7.4 and $12.7 \text{ mg N m}^{-2} \text{ d}^{-1}$ more in the eelgrass, whereas the more recent 2020 study found a small net N_2 -influx with a difference of $0.74 \text{ mg N m}^{-2} \text{ d}^{-1}$ less influx into the sediment in the eelgrass (Table 3). However, both found denitrification to be greatest within the eelgrass meadows (Aoki et al., 2020; Cole, 2011). Cole (2011) also demonstrated greater net N_2 -efflux in an 8 year restoration compared with a 3 year restoration (at a different site), with net efflux 12.6 and $7.42 \text{ mg N m}^{-2} \text{ d}^{-1}$ higher than the unvegetated control site, respectively. This suggests that denitrification increases with meadow age at restored sites (Cole, 2011). The remaining two studies displayed low net N_2 -efflux or net N_2 -influx (Kahn et al., 2023; Zarnoch et al., 2017). In the 2017 study, the sediment at Cormorant Point even showed that eelgrass had lower net N_2 -efflux than the comparable unvegetated control which is contrary to what might be expected but can be explained by ambient nutrients concentrations being sufficiently low/oligotrophic (Fig. 6). Because, when treated with an increase in NO_3^- -concentrations, all sites, including Cormorant Point, responded with the net N_2 -efflux greatly increasing in eelgrass sediments compared to the unvegetated sediments (Zarnoch et al., 2017), this demonstrates the greater potential of denitrification within the sediment of eelgrass meadows as an ecosystem function during a sudden influx of nutrients. Net N_2 -efflux increasing quickly in response to external nutrient loading (within 24 h) was also demonstrated by Kahn et al. (2023), along with a synergistic effect to acidification. This response has also been observed previously in several studies (Herbert, 1999; Kamp-Nielsen, 1992; Seitzinger, 2003) and is explained by denitrification rates responding rapidly to changes in dissolved inorganic N concentrations, thereby increasing rapidly when nutrient pulses enter the system (Zarnoch et al., 2017).

Generally, studies found a positive effect on net N_2 -flux in the eelgrass habitats, meaning greater net N_2 -efflux in eelgrass than in comparable unvegetated controls ($p = 0.02$,

$n = 10$) (Table 3), and the average ecosystem function was $14.0 \pm 4.4 \text{ mg N m}^{-2} \text{ d}^{-1}$ (mean $\pm$ propagated SE) (Fig. 6). However, as studies demonstrate, these microbial processes are sensitive to the ambient conditions of temperature, pH, oxygen and nutrients (Kahn et al., 2023), and there is evidence of seasonal importance of this ecosystem function related to these conditions, along with biotic factors (Risgaard-Petersen & Ottosen, 2000). Additionally, the current methods measure on a time scale of hours up to a day, so we cannot observe longer time-series of weeks or months, and we should be cautious in extrapolating momentary measurements.

Discussion

Applicability of Methods

Based on the literature presented in this review, current methodologies are insufficient to capture the full spectrum of nutrient-related ecosystem functions provided by eelgrass meadows. Moreover, the general lack of comparable control sites created a large knowledge gap in the literature (Fig. 5).

Sediment nutrient stocks are usually measured using sediment cores (Table 1), but there is a need for a standardized methodology that defines the appropriate sampling depth within the sediment, supported with a recommendation for the minimum number of replicates, and where to place them in the control site, as unvegetated sediments can be markedly heterogeneous (Gallop et al., 2015). Accumulation rates of nutrients are measured using the ^{210}Pb -method, which is not only expensive, easily limiting the number of replicates, but might not be very well suited for dynamic coastal sediments (Kristensen et al., 2025). However, new developments in modelling have been made to account for different anomalies, which are improving the accuracy of the method (Arias-Ortiz et al., 2018). In this review, we also suggest that the accumulation of nutrients in the sediment is not a continuous ecosystem function, but one that occurs in the developmental stages of eelgrass colonization and will reach a balance once the meadow matures (Fig. 4). There is a need for studies that focus on the developing stages of this ecosystem function after colonization of eelgrass, as well as studies that help define the parameters driving the accretion of sediments and nutrients i.e. physical forcing, meadow productivity, allochthonous and autochthonous detrital sedimentation, resuspension, and more.

The standing biomass of eelgrass has been abundantly investigated in the literature (Figs. 1 and 3), especially during summer. But there is a lack of investigation into the sum of the annual production that creates a temporary

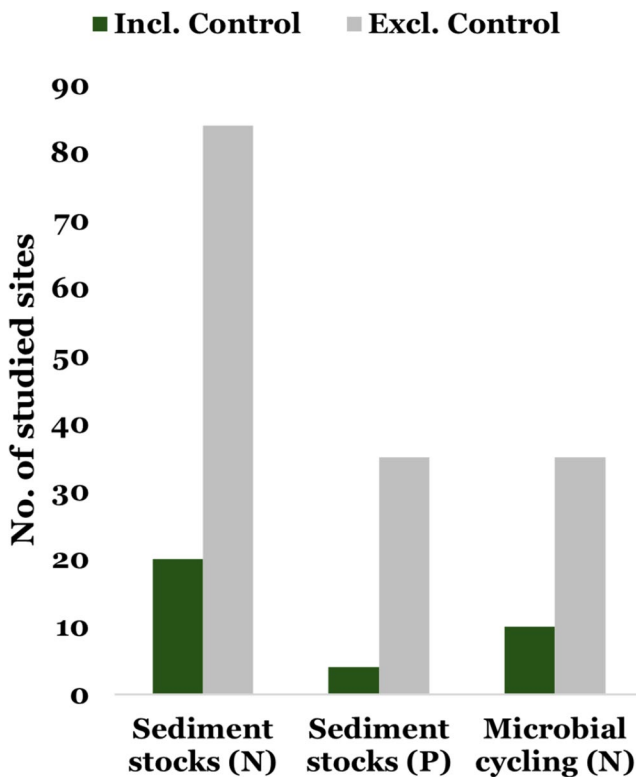


Fig. 5 The number of studied sites which either includes (green) or excludes (gray) comparable unvegetated controls to the measured eelgrass data divided into three categories of measured data: Sediment stocks of N, sediment stocks of P and microbial cycling of N

immobilization of nutrients in a detrital stock, which is a greater ecosystem function than the standing biomass alone (Sand-Jensen, 1975). However, investigating annual cycles is very time consuming and must include productivity, for which there exist several methods (Short & Duarte, 2001). Leaf marking is the most common method (Ibarra-Obando & Bourdoursque, 1994; Kentula & McIntire, 1986), but it focuses heavily on growth in individual shoots or even leaves. Alternatively, measuring the plastochrone interval can be used to scale up monitoring as it, in combination with shoot density and biomass measurements, allows for measuring the production at meadow scale (Short & Duarte, 2001). A less invasive allometric method has also been developed to measure leaf growth (Echavarría-Heras et al., 2013), this method however does not include the biomass production of the rhizomes and roots which constitutes a substantial amount of the overall biomass (Fig. 3). Studies that investigate full annual cycles of productivity across latitudinal, depth and nutrient gradients will help define the ranges of this particular ecosystem function. Additionally, measuring the minimum standing stock of biomass as part of this will give a greater insight into the proportionality of the constant contra the temporary immobilization of nutrients in eelgrass biomass.

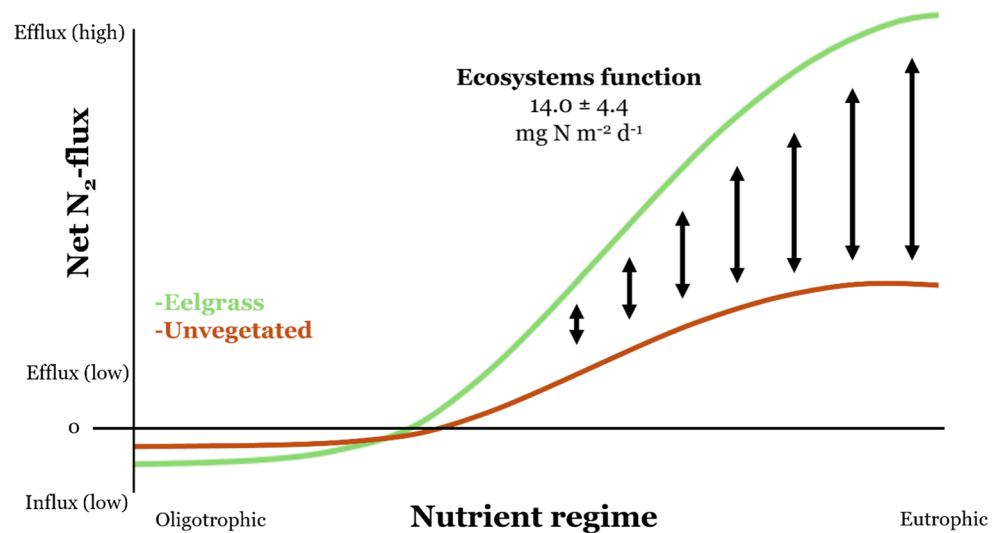
The microbially driven ecosystem functions are often investigated in the literature (Fig. 1), and the processes of denitrification and N-fixation are not exclusive to eelgrass sites. But most methods are *ex situ* (acetylene, ^{15}N -isotope, and N_2/Ar) and only provide momentary glimpses at the dynamic interaction of the microbes in sediments (Table 3). The presence of an eelgrass rhizosphere adds markedly to the complexity of these micro habitats, making it difficult to properly quantify using the acetylene, ^{15}N -isotope and $^{30}\text{N}_2$ push-pull methods. We can see that the presence of eelgrass stimulates greater denitrification (Zarnoch et al., 2017), but are unable to provide long timelines of natural events that may regulate the processes. However, advances are being made with e.g. the isotope pairing technique to account for some of the main methodological issues (Robertson et al., 2019), and the *in situ* $^{30}\text{N}_2$ push-pull method for N-fixation (Aoki & McGlathery, 2019). Some studies also call for direct comparisons between methods (N_2/Ar and isotope pairing) to intercalibrate results (Eyre et al., 2016).

Quantifying and Comparing Ecosystem Functions

The importance of nutrient-related ecosystem functions from eelgrass will vary within and between systems (McHenry et al., 2023; Namba et al., 2017). But to understand the recovery trajectories of degraded marine ecosystems, it is important to attempt to quantify the ecosystem functions lost due to drastically reduced eelgrass cover. Since sediment stocks reach a balance with the surroundings (Fig. 4), the relative contribution of this ecosystem function is low compared to that of seasonal assimilation in the biomass and microbial N turnover (Figs. 2 and 6). Assimilation in the biomass is likely the greatest contributor to immobilizing nutrients during the growth season (Risgaard-Petersen & Ottosen, 2000), whereas microbial cycling will have the greatest contribution to the permanent removal of N from the system.

Here, we present the Danish estuary “Odense Fjord” as a conceptual framework of the lost ecosystem functions provided by eelgrass. The estuary covers an area of 62 km² and was historically dominated by eelgrass meadows with ~27.3 km² of vegetated cover in the 1960s (Flindt et al., 2024). Since then, eutrophication has drastically reduced the eelgrass extent in the estuary to only ~0.6 km², resulting in a loss of 26.7 km² of eelgrass (Flindt et al., 2024). The required reduction in nutrient loading for the estuary to reach good ecological status has been estimated as an annual N loading from 2304 t N yr⁻¹ to 1713 t N yr⁻¹, according to the Water Basin Management Plan (MIM, 2023). Good ecological status in this case is defined as fulfilling three criteria: (1) Sufficiently low summer chlorophyll A concentrations in the water column, (2) Presence of rooted vegetation at a target depth based on a reference state and (3) Benthic

Fig. 6 Conceptual diagram of net N_2 -flux in eelgrass (light green line) and unvegetated (red line) sediments going from an oligotrophic to a eutrophic nutrient regime. Ecosystem function (black double headed arrows) is the mean of means $\pm$ propagated SE of the difference between eelgrass and unvegetated controls ($n = 10$)



invertebrate biodiversity. But the reduction goal is based on a reference state of N loading and does not account for the absence of eelgrass coverage in the system (MIM, 2023). Therefore, the lost ecosystem functions are not included in the calculation. From the results presented in this review, the ecosystem functions lost along with the eelgrass coverage would amount to a rough estimate of:

- Standing biomass (Fig. 3): $5.2 \text{ g N m}^{-2} \times 26.7 \text{ km}^2 = 139 \text{ T N}$, which corresponds to a yearly production of eelgrass (Sand-Jensen, 1975): $139 \text{ T} \times 2.5 = 347 \text{ T N yr}^{-1}$.
- Sediment stock (Fig. 4): $19.9 \text{ g N m}^{-2} \times 26.7 \text{ km}^2 = 531 \text{ T N}$.
- gNet N_2 efflux of (Fig. 6): $14.0 \text{ mg N m}^{-2} \text{ d}^{-1} \times 26,7 \text{ km}^2 = 136 \text{ T N yr}^{-1}$.

All of which are not accounted for in the water body's nutrient reduction goal (MIM, 2023). Even though these are rough estimates which will vary from site to site, the general magnitude of these are important for decision makers to be aware of, as the estimated reduction goal is likely not enough to reach good ecological status due to the fact that the recovery trajectory has shifted (Duarte et al., 2008; Flindt et al., 2024). Even if the external nutrient loading is reduced to a natural baseline level, the bioavailable nutrients which previously would have been assimilated, immobilized, buried and exported in and from eelgrass will be available for opportunistic algae, and thus the lack of eelgrass can be the limiting factor to achieving good ecological conditions (Pedersen et al., 2013). In Odense Fjord today, the remaining eelgrass is stable but limited to a few narrow bands at a depth of 1–2 m as the lower depth limit is controlled by light availability and sediment conditions, while the upper depth limit is controlled by the hydrodynamic stress from waves (Flindt et al., 2016). In cases like this, where the distribution

and coverage are extremely limited, the recovery of eelgrass can be limited by a lack of propagules, but also due to the lack of the eelgrass sediment stabilizing function, which keeps the estuary in a turbid state (de Boer, 2007; Hansen & Reidenbach, 2012). Here, active restoration of eelgrass can be a supplemental measure to help accelerate the return of ecosystem functions and achieve good ecological conditions (Flindt et al., 2024; Lillebø et al., 2011; Orth et al., 2020), but it also provides unique research opportunities, such as performing BACI studies, which provide the highest quality assessment (Christie et al., 2019; Reynolds et al., 2016). This is because the BACI study design includes visiting a site before impact, which in this case could be transplantation of eelgrass, after impact, and comparing the impacted site to a meaningful control site. In addition, if it is possible to compare to a natural eelgrass meadow, this “positive” control can help evaluate the development and success of a restoration and the associated ecosystem functions. This is not only the case for eelgrass but for all seagrasses where successful restoration efforts are taking place, such as *Cymodocea nodosa*, *Posidonia sp.*, *Zostera muelleri*, and *Zostera noltii*. (Da Ros et al., 2020; Kendrick et al., 2025; Matheson et al., 2022; Oliveira et al., 2025).

Conclusion

In this review, we found that the nutrient-related ecosystem functions in eelgrass are not yet sufficiently quantified, and that current methods do not fully encompass the underlying mechanisms. This can be attributed to a low number of published studies on specific topics, large methodological variability, a lack of control sites, a high spatial variation in rates and stock sizes, and a need for technological advancement. This is unfortunate, as eelgrass provides essential ecosystem

functions, and its nutrient-buffering capacities are vital to ensure ecological stability. This contribution was evident in the available data, as we found that eelgrass can retain $19.9 \pm 89.1 \text{ g N m}^{-2}$ and $4.8 \pm 2.3 \text{ g P m}^{-2}$ through nutrient burial and sedimentary stocks, immobilize $5.1 \pm 2.5 \text{ g N m}^{-2}$ and $0.6 \pm 0.2 \text{ g P m}^{-2}$ by assimilation in the standing biomass, and export $14.0 \pm 17.6 \text{ mg N m}^{-2} \text{ d}^{-1}$ through microbially driven nutrient cycling. However, there is a need for further quantification of nutrient-related ecosystem functions with a greater temporal and spatial resolution, while accounting for the conceptual mechanisms driving the processes. Here, restored eelgrass meadows allow for unique possibilities to perform BACI-studies, which provide accurate quantification of the ecosystem functions related to the age of the meadow. Continued research in natural populations will also help us understand the environmental drivers of the ecosystem functions and assess the success and development of restorations. Measurements across seagrass systems in diverse and novel environmental contexts will help set upper and lower bounds on the rates and scale of the ecosystem functions. Thus, there are many opportunities for different research to contribute to our further understanding of these ecosystem functions. This will provide us with further insight into the loss of eelgrass ecosystem functions, how it potentially affects the recovery trajectory of degraded water bodies, and how the absence of this vital habitat can be the limiting factor in achieving good ecological conditions.

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Data Availability The data that supports the findings of this study are available in the supplementary material of this article.

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